# Baseball Statcast
# 2025 Out-of-Sample Forecast Report

---

**Training period:** 2019–2024 | **Validation period:** 2025

Models were trained exclusively on 2019–2024 data, then applied to 2025 players who were never seen during training. This report measures true out-of-sample accuracy using MAE, RMSE, $R^2$, and Percent Error — a more rigorous test than cross-validated $R^2$ on historical data.

## Part 1 — Training Model Summary (2019–2024)

Models were selected and tuned using 2019–2024 data only. The best model per target (by held-out test $R^2$) was retained for 2025 deployment.

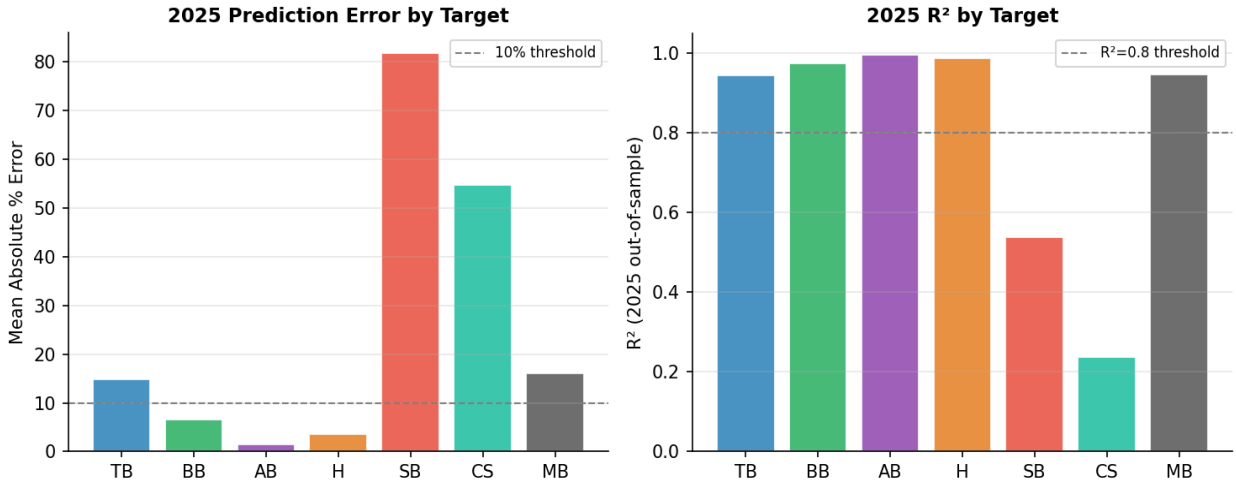
| Target | Best Model | Train $R^2$ | CV $R^2$ | Train RMSE |
|---|---|---|---|---|
| TB | Linear Regression | 0.9541 | 0.9484±0.0126 | 17.1072 |
| BB | Lasso Regression | 0.8929 | 0.8953±0.0149 | 7.6481 |
| AB | Lasso Regression | 0.9864 | 0.9888±0.0013 | 19.2832 |
| H | Lasso Regression | 0.9569 | 0.9587±0.0067 | 9.9207 |
| SB | Poisson GLM (5 features) | 0.4099 | 0.405 | 7.95 |
| CS | Neg. Binomial GLM (5 features) | 0.2863 | 0.2804 | 2.2468 |

## Part 2 — 2025 Out-of-Sample Validation

The trained models were applied to 2025 batters. The table below shows how well the predictions match the actual 2025 statistics. $R^2$ here is *not* a training metric — it reflects genuine predictive accuracy on unseen data.

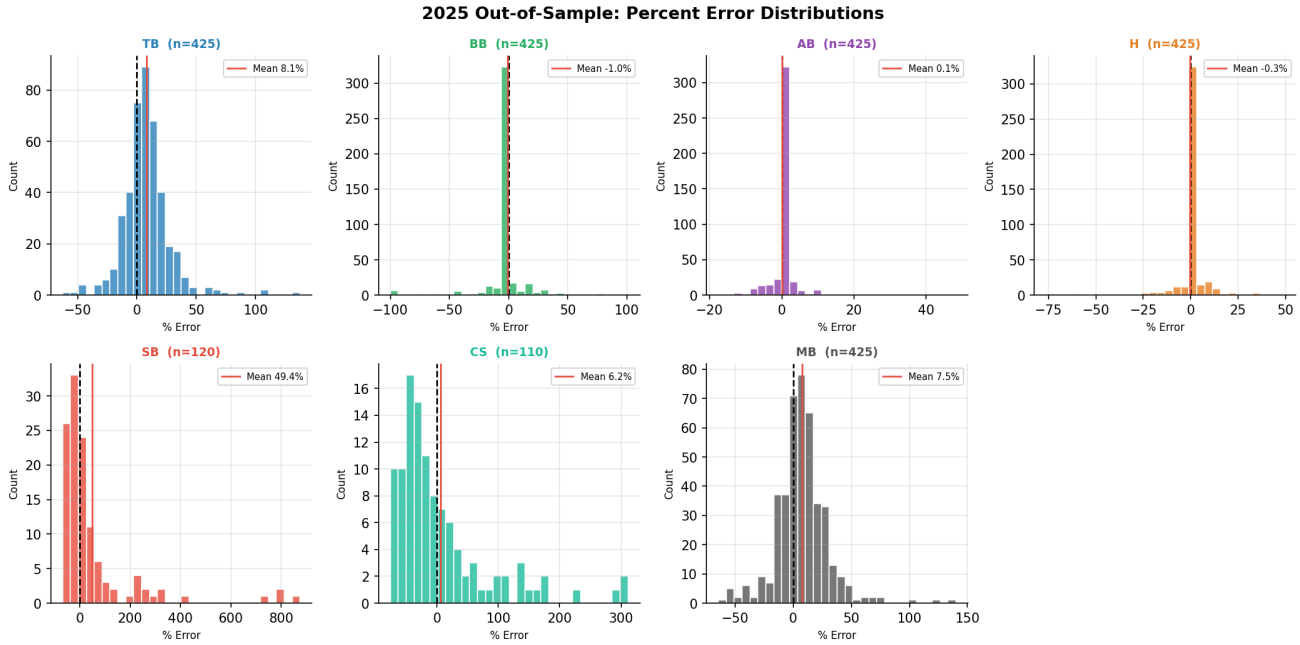
| Target | n | $R^2$ (2025) | MAE | RMSE | Mean APE% | ±5% | ±10% | ±20% |
|---|---|---|---|---|---|---|---|---|
| TB | 425 | 0.9436 | 15.224 | 19.597 | 14.8% | 24.9% | 46.8% | 76.9% |
| BB | 425 | 0.9737 | 1.425 | 3.692 | 6.7% | 78.1% | 81.2% | 90.1% |
| AB | 425 | 0.9961 | 4.301 | 10.508 | 1.5% | 88.9% | 96.5% | 99.1% |
| H | 425 | 0.9868 | 2.314 | 5.372 | 3.6% | 79.8% | 88.7% | 95.3% |
| SB | 122 | 0.5384 | 5.174 | 7.219 | 81.9% | 9.8% | 16.4% | 30.3% |
| CS | 122 | 0.2369 | 1.854 | 2.421 | 54.8% | 3.3% | 12.3% | 21.3% |
| MB | 425 | 0.9449 | 5.238 | 6.953 | 16.1% | 22.4% | 43.8% | 74.4% |

### Summary Overview




Left: Mean Absolute Percent Error per target. Lower is better. Right: Out-of-sample $R^2$ per target. Higher is better.

### Percent Error Distributions

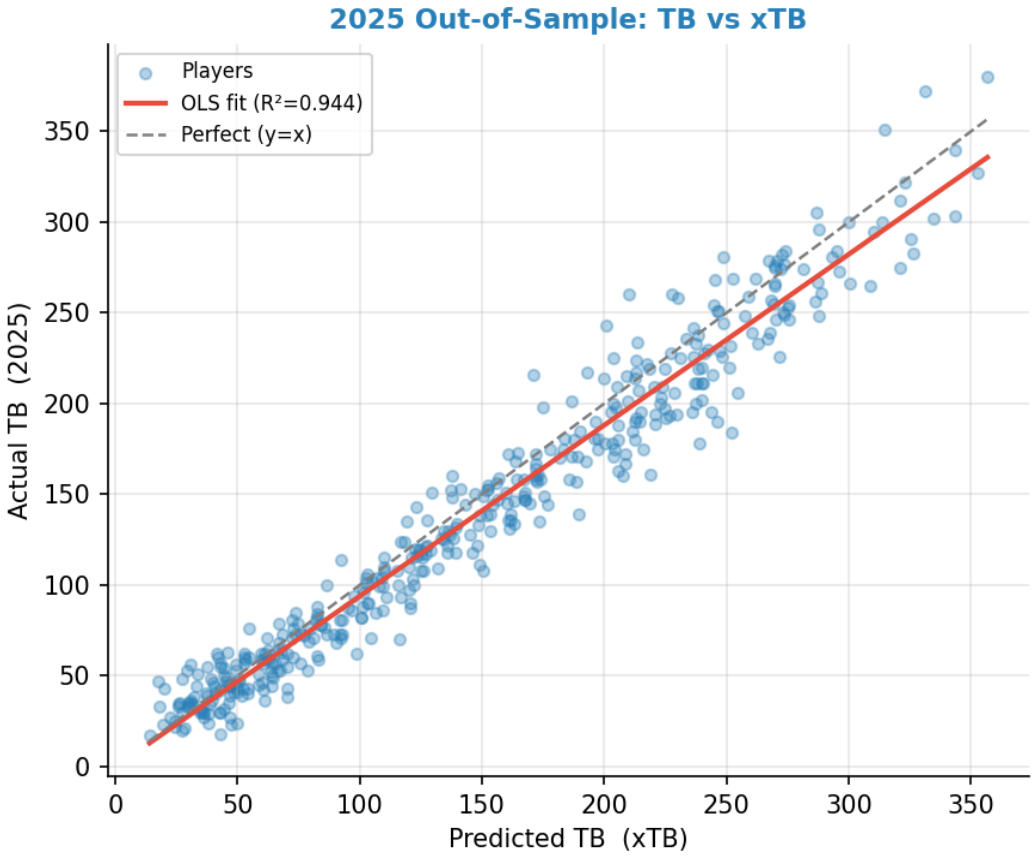


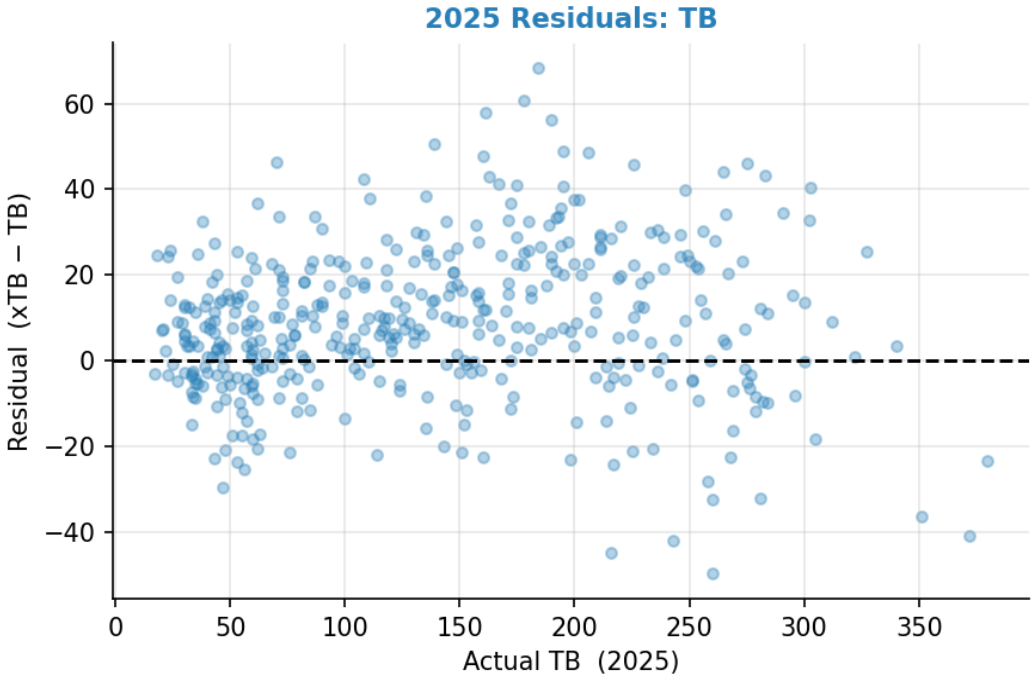

Percent error distributions for each target in 2025. A distribution centred on zero indicates unbiased predictions.

## Per-Target: Predicted vs Actual (2025)

### Target: TB

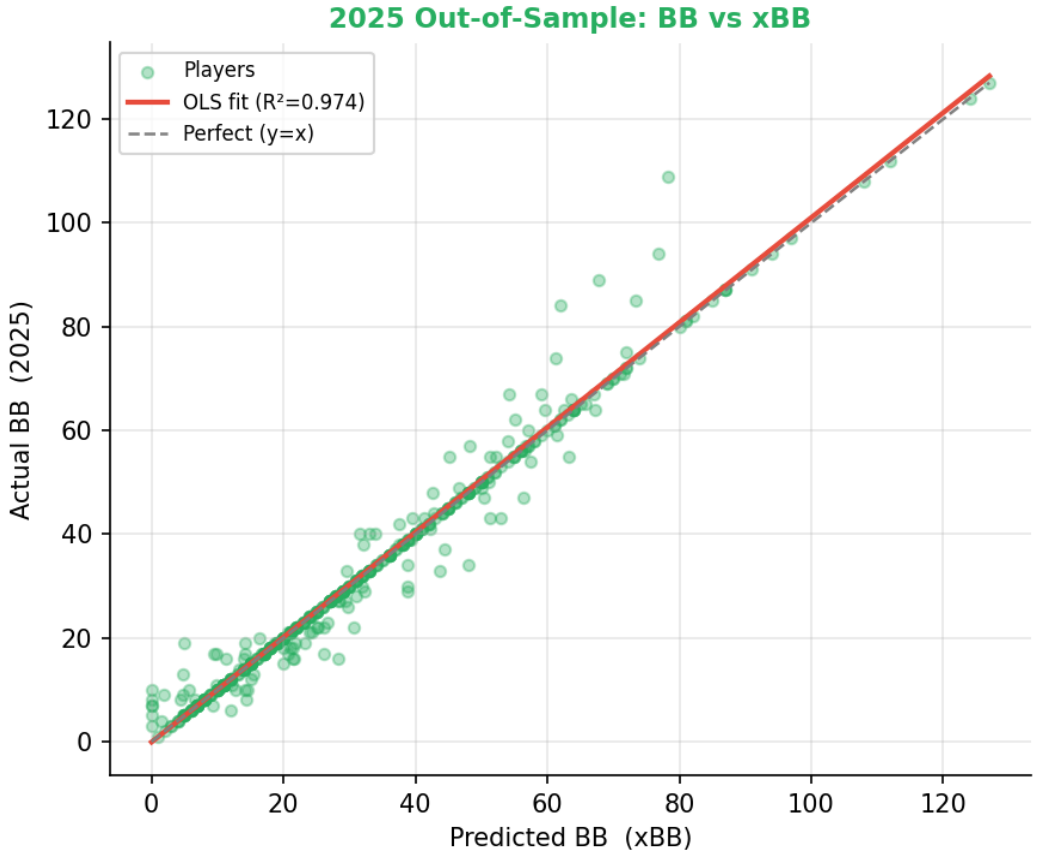
n=425 | R²=0.9436 | MAE=15.224 | RMSE=19.597 | Mean APE=14.76%

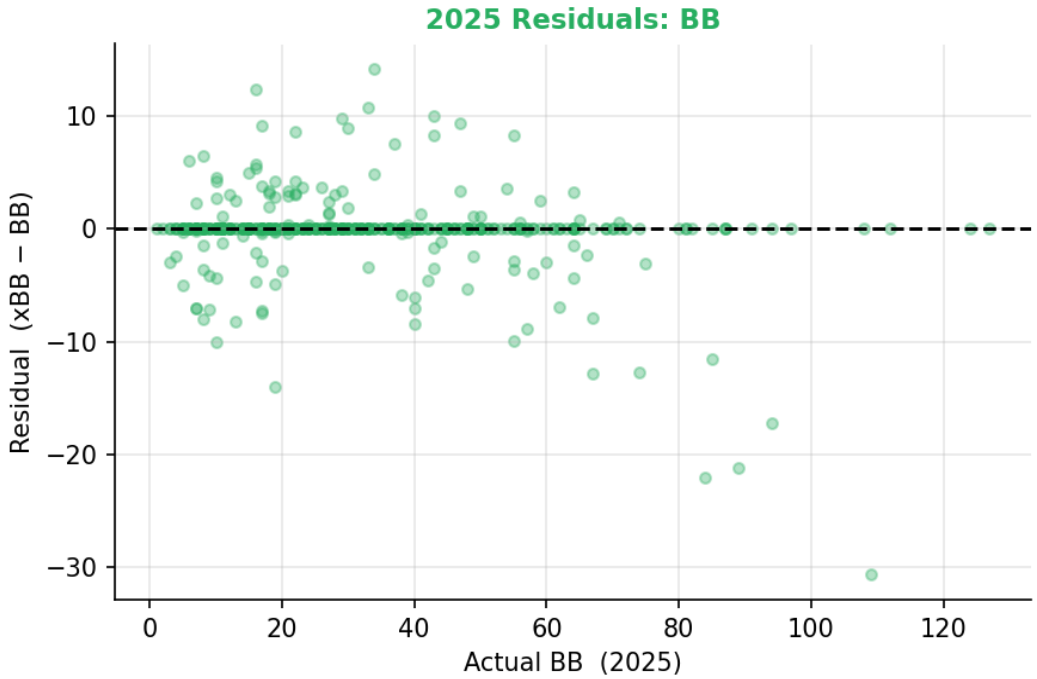


## Target: BB

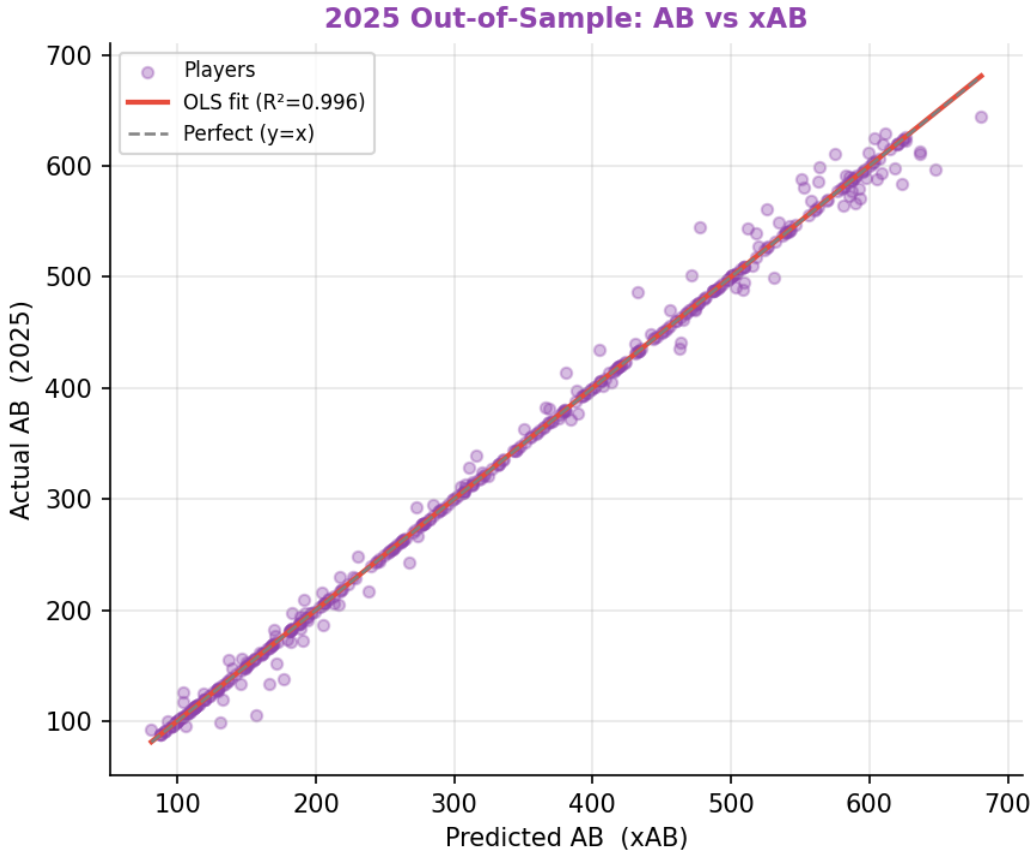
n=425 | R²=0.9737 | MAE=1.425 | RMSE=3.692 | Mean APE=6.65%

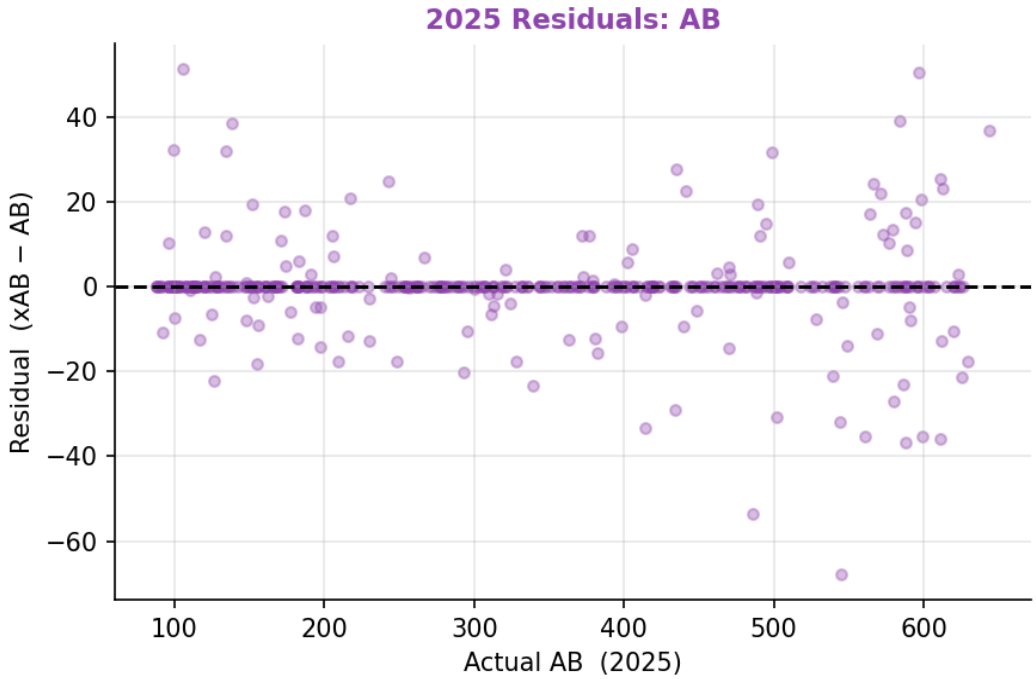


## Target: AB

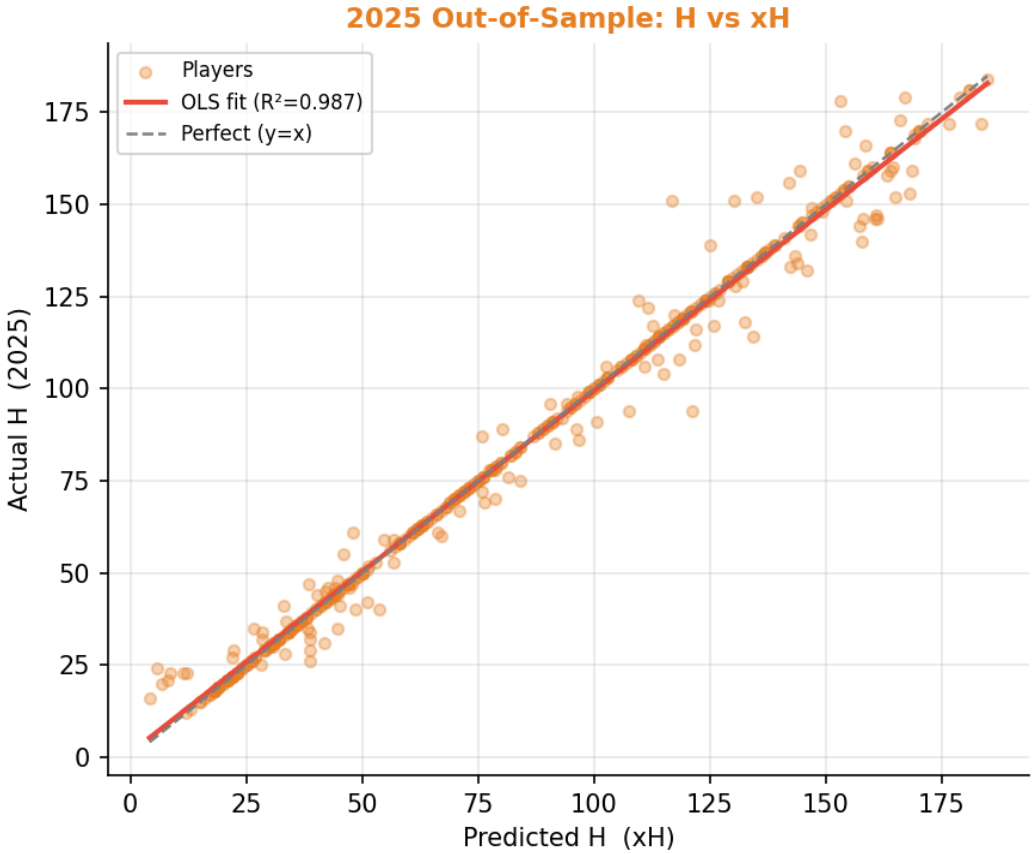
n=425 | R²=0.9961 | MAE=4.301 | RMSE=10.508 | Mean APE=1.48%

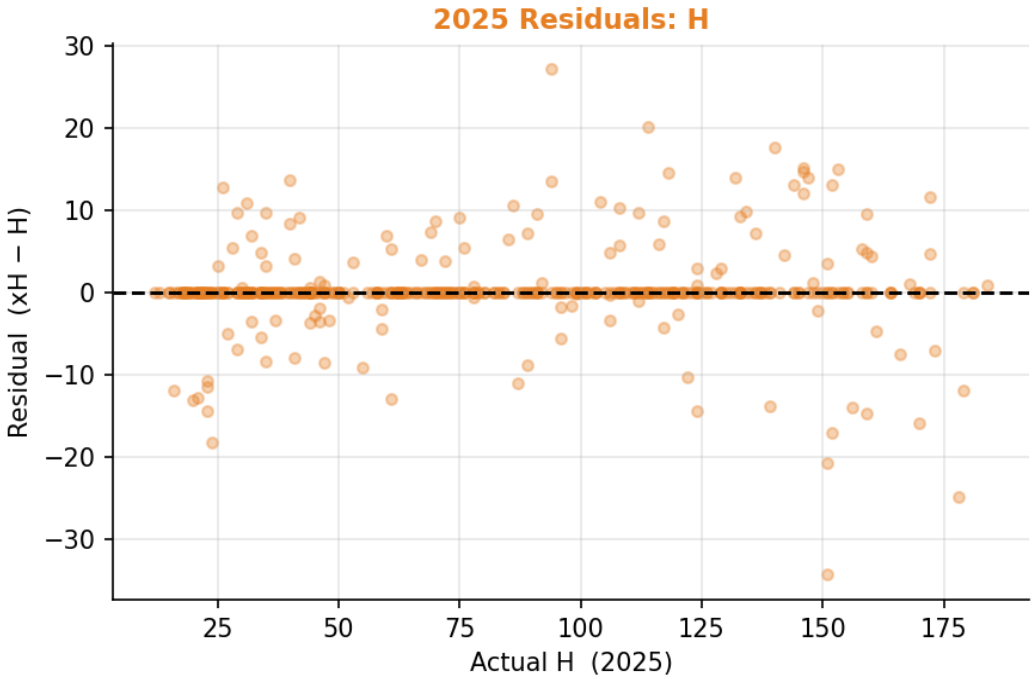


## Target: H

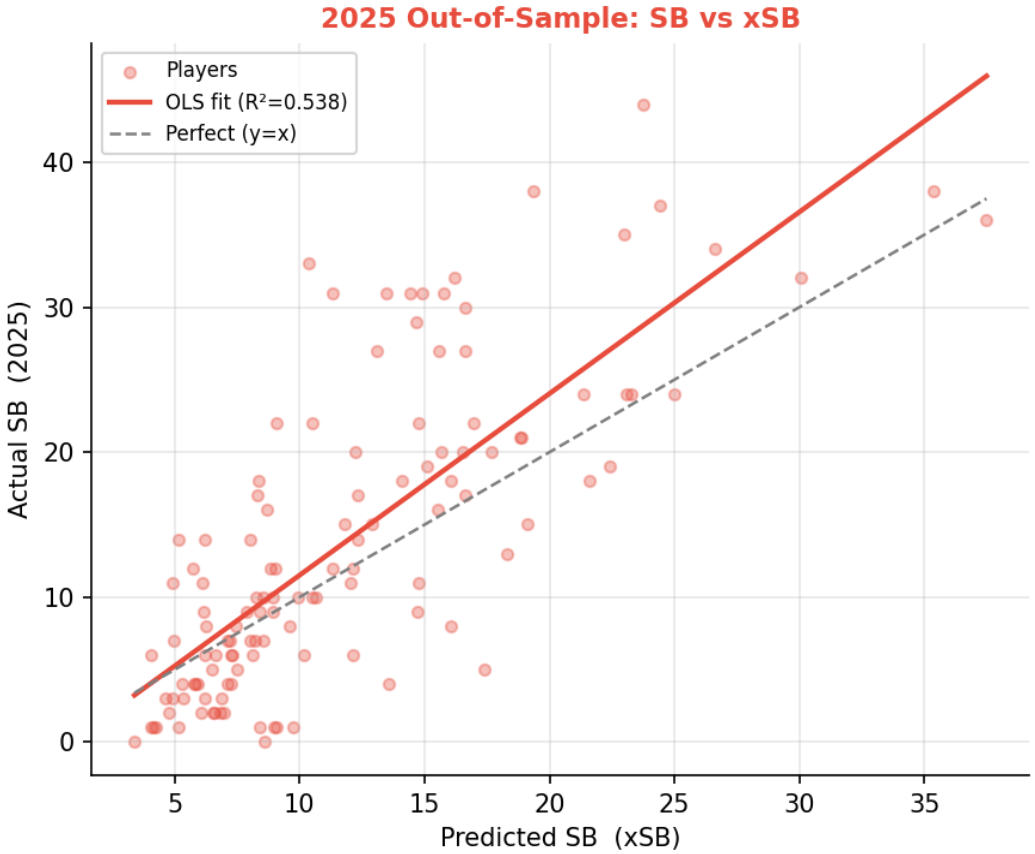
n=425 | R²=0.9868 | MAE=2.314 | RMSE=5.372 | Mean APE=3.65%

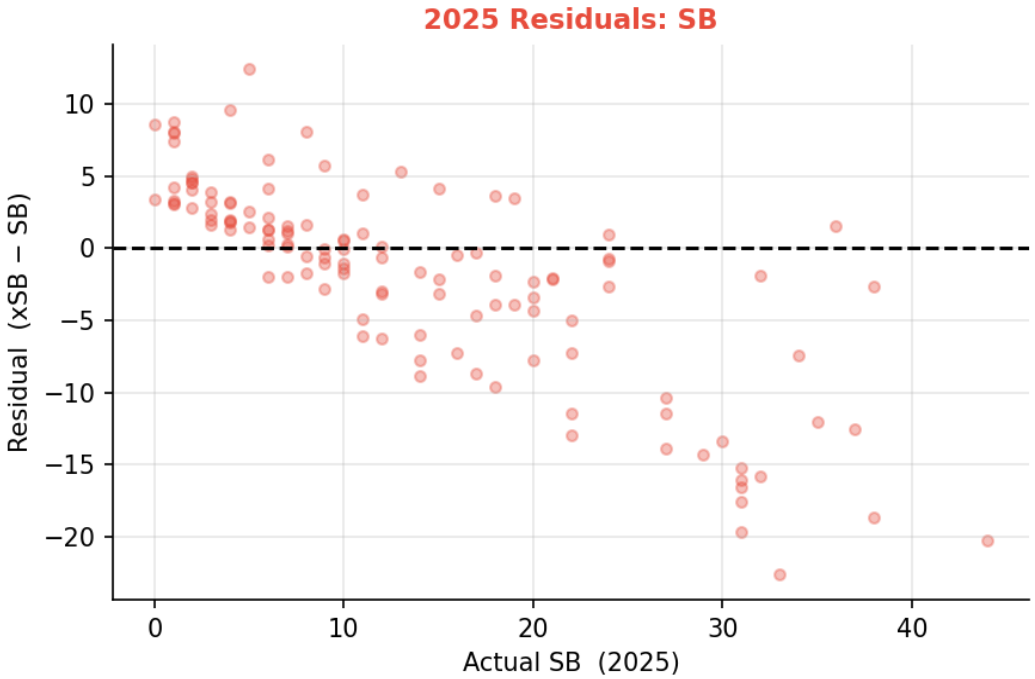


## Target: SB

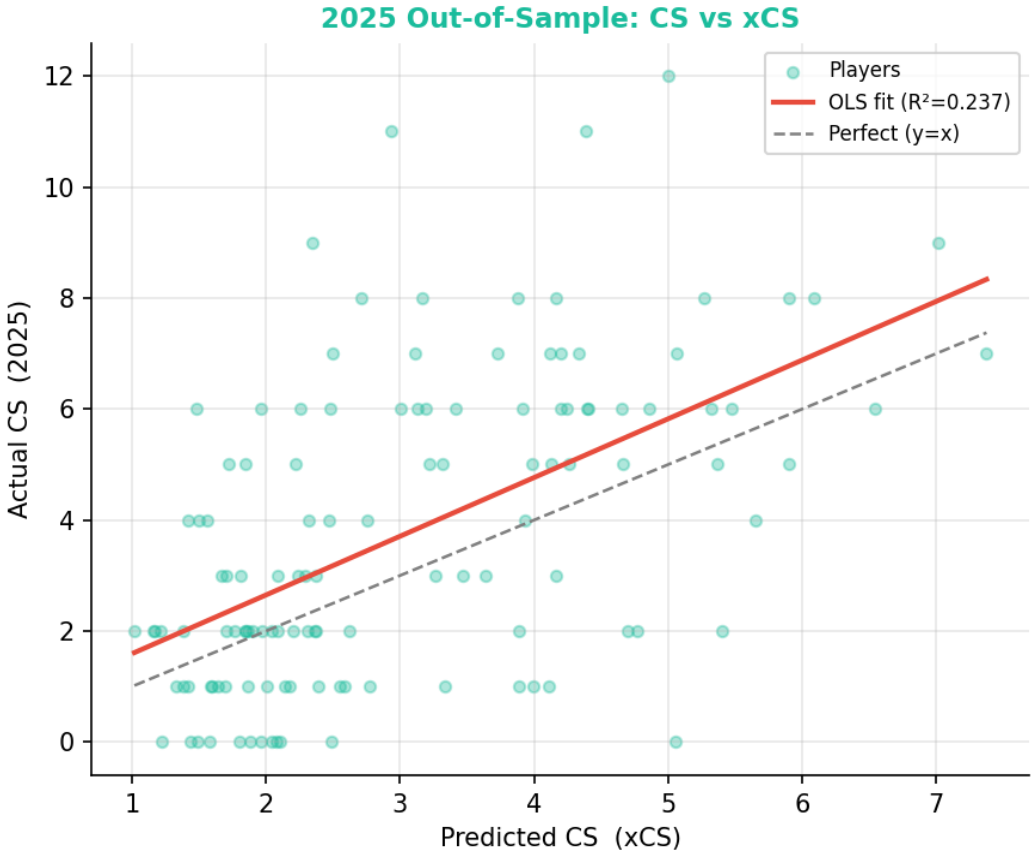
n=122 | $R^2$=0.5384 | MAE=5.174 | RMSE=7.219 | Mean APE=81.88%

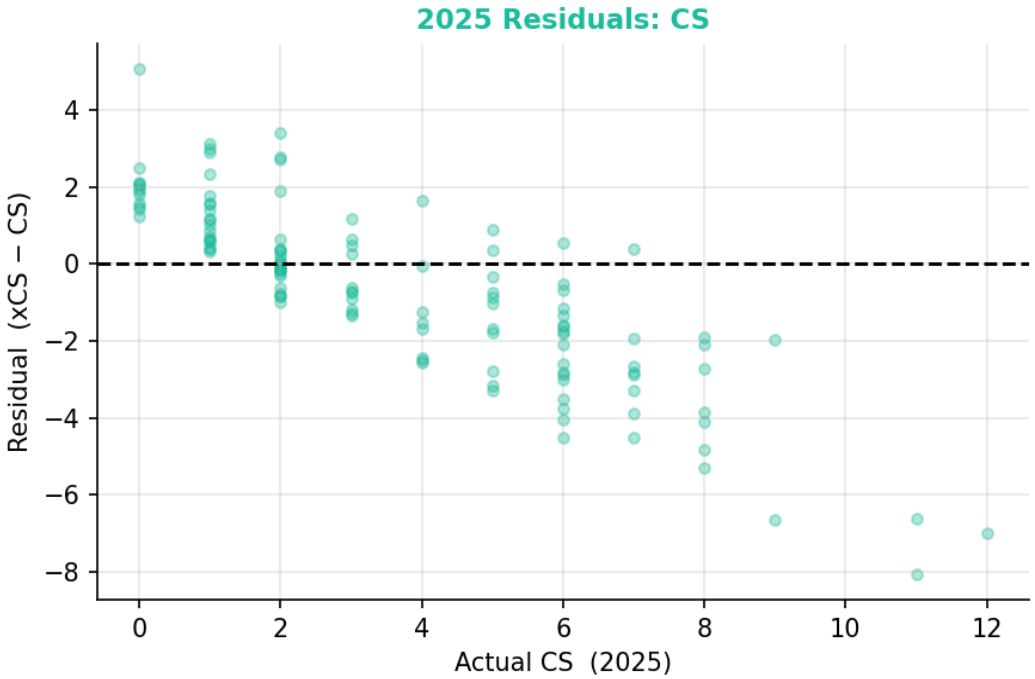


## Target: CS

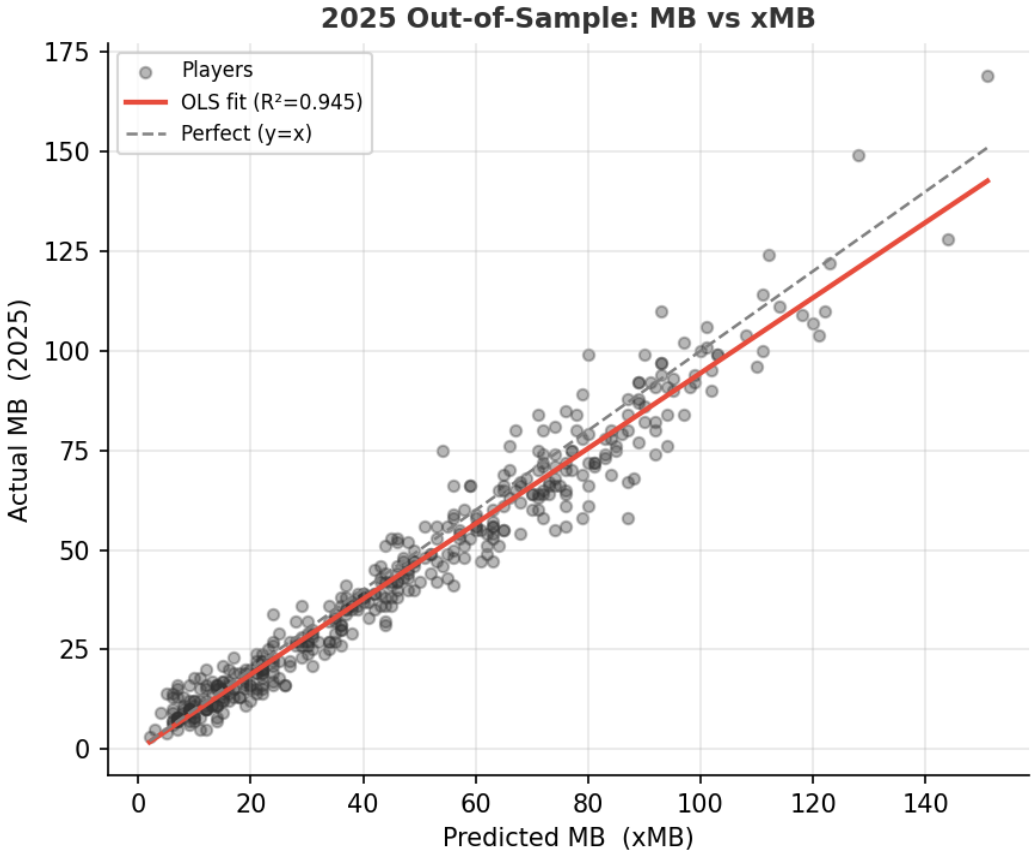
n=122 | R²=0.2369 | MAE=1.854 | RMSE=2.421 | Mean APE=54.75%

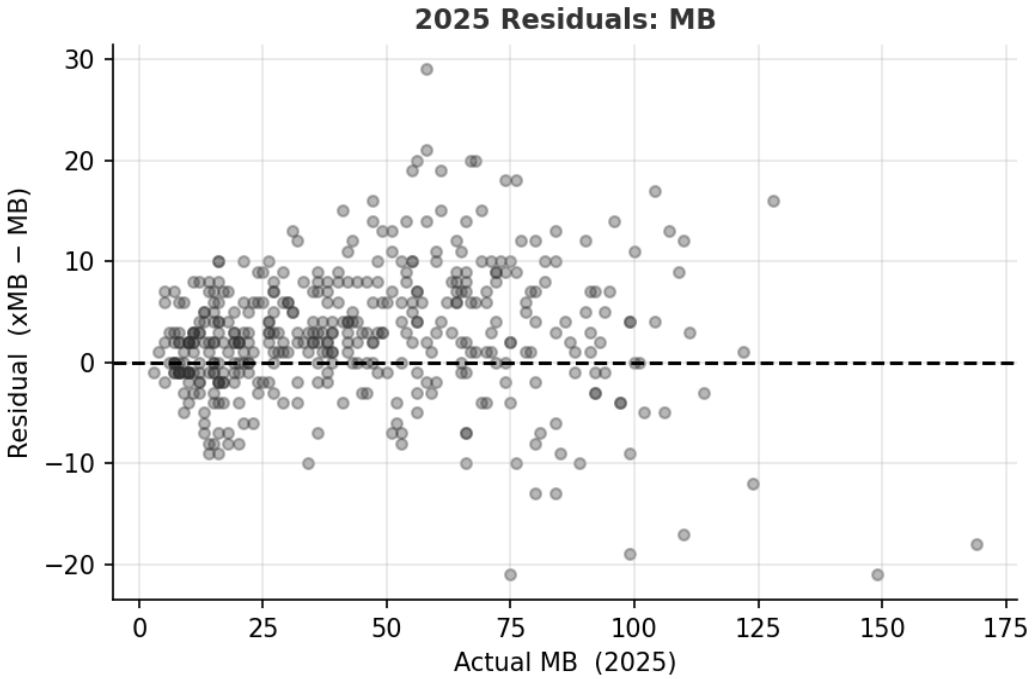


## Target: MB

n=425 | R²=0.9449 | MAE=5.238 | RMSE=6.953 | Mean APE=16.12%



## Part 3 — 2025 Player-Level Predictions vs Actuals

First 30 rows of the 2025 deployment. Full results saved to **expected_2025_stats.csv**.

| Name | Sea son | TB | xTB | BB | xBB | AB | xAB | H | xH | SB | xSB | CS | xCS | MB | xM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Trea Turner | 2025.0 | 269.0 | 262.04 | 43.0 | 41.34 | 589.0 | 597.66 | 179.0 | 167.14 | 36.0 | 37.49 | 7.0 | 7.37 | 99.0 | 90 |
| Victor Scott II | 2025.0 | 118.0 | 139.29 | 42.0 | 37.45 | 398.0 | 388.55 | 86.0 | 96.65 | 34.0 | 26.6 | 4.0 | 5.65 | 40.0 | 49 |
| Bobby Witt Jr. | 2025.0 | 312.0 | 321.11 | 49.0 | 46.55 | 623.0 | 625.75 | 184.0 | 184.84 | 38.0 | 35.37 | 9.0 | 7.02 | 111.0 | 114 |
| Byron Buxton | 2025.0 | 269.0 | 252.7 | 41.0 | 42.3 | 488.0 | 486.63 | 129.0 | 131.94 | 24.0 | 23.09 | 0.0 | 5.06 | 92.0 | 89 |
| Derek Hill | 2025.0 | 44.0 | 46.68 | 9.0 | 4.83 | 134.0 | 146.05 | 29.0 | 38.63 | 7.0 | 8.23 | 0.0 | 2.04 | 13.0 | 15 |
| Johan Rojas | 2025.0 | 44.0 | 47.0 | 13.0 | 4.73 | 152.0 | 171.47 | 34.0 | 38.78 | 12.0 | 9.02 | 2.0 | 2.2 | 14.0 | 13 |
| andon Lockridge | 2025.0 | 40.0 | 40.88 | 8.0 | 0.0 | 134.0 | 165.95 | 31.0 | 41.86 | 10.0 | 8.92 | 1.0 | 2.18 | 12.0 | 12 |
| Eli White | 2025.0 | 104.0 | 102.35 | 11.0 | 9.77 | 256.0 | 255.85 | 60.0 | 66.97 | 10.0 | 10.64 | 1.0 | 2.58 | 29.0 | 31 |
| Jake McCarthy | 2025.0 | 71.0 | 81.3 | 10.0 | 14.5 | 206.0 | 213.02 | 42.0 | 51.19 | 6.0 | 10.16 | 0.0 | 2.49 | 18.0 | 25 |
| Bryce Teodosio | 2025.0 | 42.0 | 43.02 | 5.0 | 0.0 | 138.0 | 176.43 | 28.0 | 33.37 | 7.0 | 8.04 | 1.0 | 2.01 | 10.0 | 9. |
| Corbin Carroll | 2025.0 | 305.0 | 286.65 | 67.0 | 54.21 | 564.0 | 581.03 | 146.0 | 161.18 | 32.0 | 30.06 | 6.0 | 6.55 | 106.0 | 10. |
| handler Simpson | 2025.0 | 143.0 | 122.94 | 20.0 | 16.28 | 414.0 | 380.71 | 122.0 | 111.7 | 44.0 | 23.76 | 12.0 | 5.0 | 51.0 | 44 |
| Tyler Fitzgerald | 2025.0 | 71.0 | 72.07 | 17.0 | 9.77 | 217.0 | 237.76 | 47.0 | 47.83 | 9.0 | 8.92 | 4.0 | 2.31 | 19.0 | 17 |
| Denzel Clarke | 2025.0 | 55.0 | 37.44 | 6.0 | 12.06 | 148.0 | 140.02 | 34.0 | 28.53 | 6.0 | 7.28 | 0.0 | 1.89 | 15.0 | 11 |
| e Crow-Armstrong | 2025.0 | 284.0 | 274.06 | 29.0 | 38.76 | 591.0 | 583.02 | 146.0 | 158.1 | 35.0 | 22.99 | 8.0 | 5.27 | 82.0 | 90 |
| Brenton Doyle | 2025.0 | 189.0 | 220.76 | 30.0 | 38.86 | 502.0 | 471.34 | 117.0 | 125.72 | 18.0 | 16.07 | 2.0 | 3.89 | 55.0 | 74 |
| Edmundo Sosa | 2025.0 | 114.0 | 92.17 | 10.0 | 0.0 | 243.0 | 267.76 | 67.0 | 70.91 | 1.0 | 8.42 | 1.0 | 2.14 | 34.0 | 24 |
| Michael Helman | 2025.0 | 45.0 | 58.49 | 7.0 | 0.0 | 99.0 | 131.18 | 23.0 | 12.22 | 4.0 | 5.92 | 0.0 | 1.58 | 14.0 | 6. |
| Otto Kemp | 2025.0 | 81.0 | 82.24 | 12.0 | 15.05 | 197.0 | 192.04 | 46.0 | 42.51 | 2.0 | 6.82 | 2.0 | 1.85 | 22.0 | 22 |
| Jonatan Clase | 2025.0 | 30.0 | 35.84 | 10.0 | 14.17 | 100.0 | 92.67 | 21.0 | 8.18 | 3.0 | 6.2 | 1.0 | 1.64 | 9.0 | 8. |
| Myles Straw | 2025.0 | 98.0 | 101.08 | 19.0 | 14.12 | 267.0 | 273.79 | 70.0 | 78.64 | 12.0 | 11.32 | 1.0 | 2.78 | 33.0 | 35 |
| Cam Smith | 2025.0 | 158.0 | 173.79 | 43.0 | 39.44 | 441.0 | 463.42 | 104.0 | 114.98 | 8.0 | 16.04 | 1.0 | 4.11 | 49.0 | 55 |
| David Hamilton | 2025.0 | 59.0 | 68.34 | 13.0 | 15.47 | 177.0 | 171.07 | 35.0 | 44.74 | 22.0 | 9.07 | 6.0 | 2.25 | 16.0 | 24 |
| Jacob Young | 2025.0 | 93.0 | 116.54 | 27.0 | 29.34 | 324.0 | 319.91 | 75.0 | 84.12 | 15.0 | 11.81 | 11.0 | 2.93 | 26.0 | 36 |
| Lane Thomas | 2025.0 | 34.0 | 38.65 | 14.0 | 13.33 | 125.0 | 118.44 | 20.0 | 6.87 | 4.0 | 5.76 | 1.0 | 1.59 | 9.0 | 6. |
| Will Benson | 2025.0 | 100.0 | 115.79 | 16.0 | 21.68 | 230.0 | 217.08 | 52.0 | 51.38 | 2.0 | 6.96 | 2.0 | 1.85 | 27.0 | 35 |
| Tyrone Taylor | 2025.0 | 99.0 | 107.88 | 16.0 | 13.92 | 310.0 | 308.15 | 69.0 | 76.36 | 12.0 | 8.86 | 2.0 | 2.38 | 27.0 | 32 |
| Oswald Peraza | 2025.0 | 61.0 | 82.38 | 17.0 | 14.09 | 244.0 | 245.99 | 40.0 | 48.41 | 9.0 | 7.9 | 0.0 | 2.11 | 15.0 | 21 |
| Matt McLain | 2025.0 | 175.0 | 203.98 | 55.0 | 52.17 | 510.0 | 515.59 | 112.0 | 121.72 | 18.0 | 21.6 | 2.0 | 5.4 | 55.0 | 65 |

| Name | Sea son | TB | xTB | BB | xBB | AB | xAB | H | xH | SB | xSB | CS | xCS | MB | xM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ackson Chourio | 2025.0 | 254.0 | 244.78 | 30.0 | 31.89 | 549.0 | 534.9 | 148.0 | 149.15 | 21.0 | 18.82 | 7.0 | 4.33 | 78.0 | 79 |

# Part 4 — MB vs xMB: Bayesian Comparison (2025 Data)

The Bayesian comparison now uses **2025 actual MB** vs **xMB predicted by the 2019–2024 model**. This is a true out-of-sample test of whether xMB carries meaningful predictive signal about MB in an unseen season.

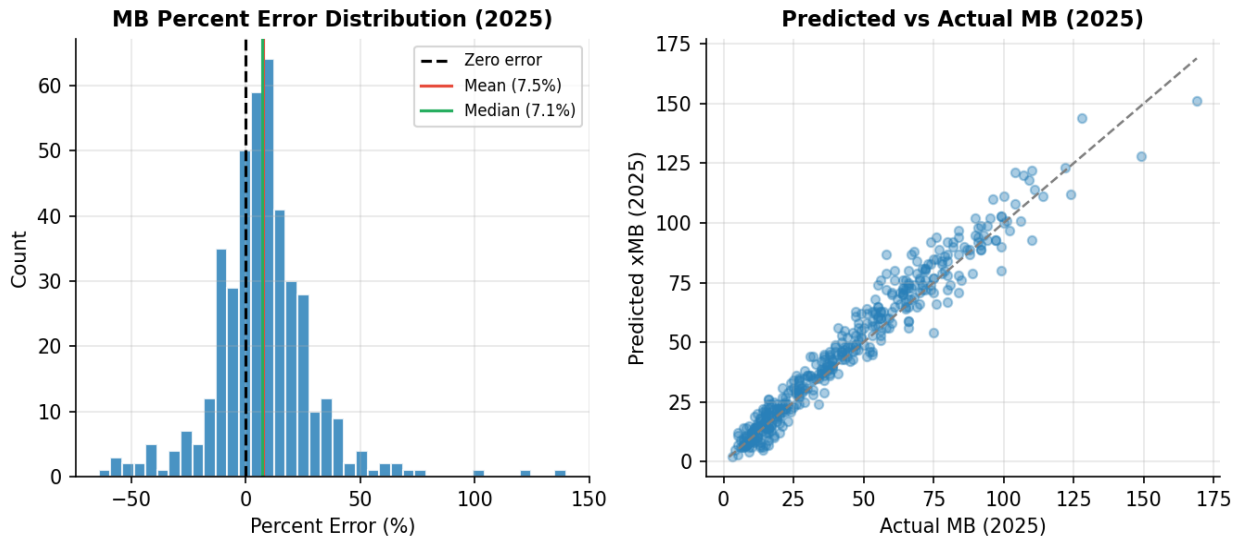
## xMB Component Selection

Each of the 6 component stats is evaluated against the $R^2$ threshold of **0.5**. If a component's best training $R^2$ meets the threshold, the *expected* (x<stat>) value is used in the xMB formula; otherwise the *actual observed* stat is used.

| Component | Training $R^2$ | Threshold (≥0.5) | Used in xMB |
|---|---|---|---|
| H | 0.9569 | ✓ Pass | xH |
| BB | 0.8929 | ✓ Pass | xBB |
| CS | 0.2863 | ✗ Fail | CS |
| TB | 0.9541 | ✓ Pass | xTB |
| SB | 0.4099 | ✗ Fail | SB |
| AB | 0.9864 | ✓ Pass | xAB |

Components marked ✗ Fail use the real observed stat in the xMB formula.

| Metric | Value |
|---|---|
| Players analysed | 425 |
| Mean Percent Error | 7.54% (+ = over-predict) |
| Median Percent Error | 7.14% |
| Mean Abs. Percent Error | 16.12% |
| Median Abs. Percent Error | 11.54% |
| Std Dev of Percent Error | 21.83% |
| Min / Max Percent Error | -64.29% / 140.0% |
| Players within ±5% | 22.4% |
| Players within ±10% | 43.8% |
| Players within ±20% | 74.4% |




Left: distribution of MB percent error in 2025. Right: xMB vs MB scatter.

## Top 5 Over-Predicted (xMB >> MB)

| Name | Season | MB | xMB | % Error |
|---|---|---|---|---|
| Hunter Feduccia | 2025 | 5.0 | 12.0 | 140.0% |
| Ben Rortvedt | 2025 | 5.0 | 11.0 | 120.0% |
| Cooper Hummel | 2025 | 7.0 | 14.0 | 100.0% |
| Matt Vierling | 2025 | 8.0 | 14.0 | 75.0% |
| Jack Suwinski | 2025 | 11.0 | 19.0 | 72.7% |

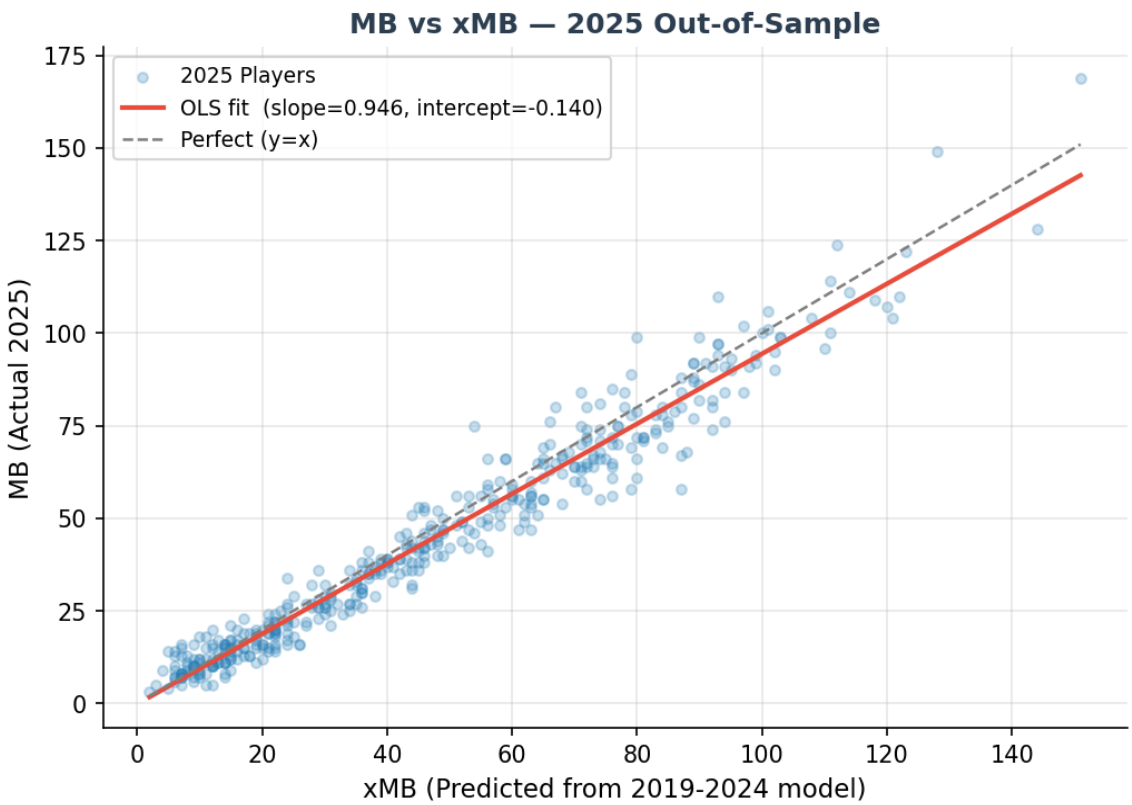
## Top 5 Under-Predicted (xMB << MB)

| Name | Season | MB | xMB | % Error |
|---|---|---|---|---|
| Adrian Del Castillo | 2025 | 14.0 | 5.0 | -64.3% |
| Michael Helman | 2025 | 14.0 | 6.0 | -57.1% |
| Joey Loperfido | 2025 | 16.0 | 7.0 | -56.2% |
| Braxton Fulford | 2025 | 9.0 | 4.0 | -55.6% |
| Marcelo Mayer | 2025 | 13.0 | 6.0 | -53.9% |

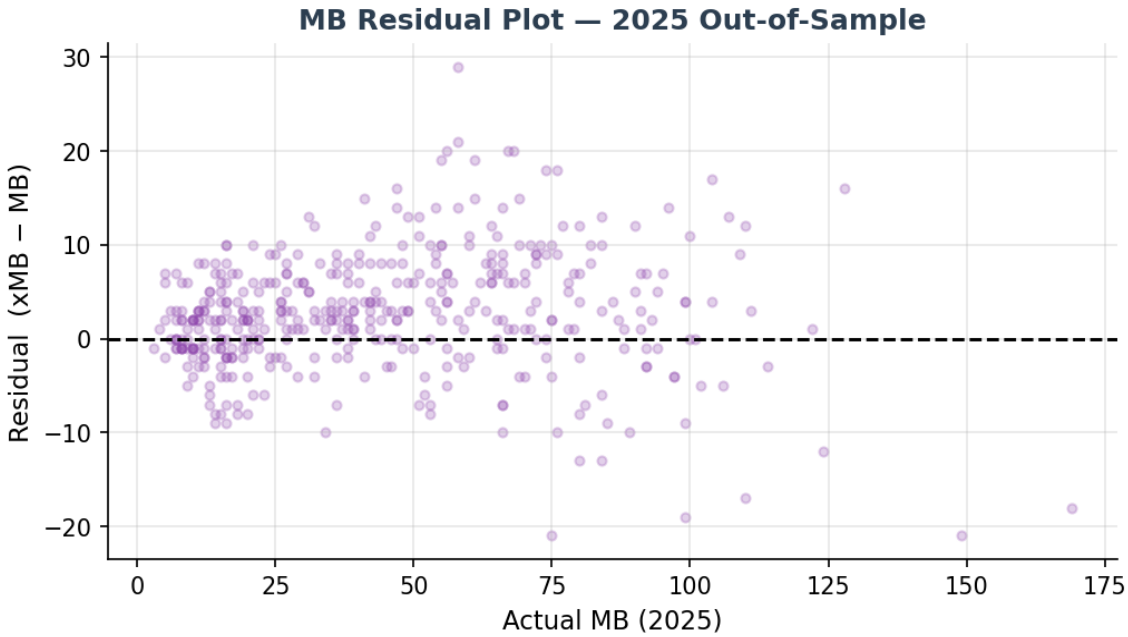
## Bayesian Model Comparison (2025)

| Statistic | Value |
|---|---|
| BIC — Model 0 (Null) | 2886.52 |
| BIC — Model 1 (xMB predictor) | 1561.64 |
| Delta BIC (BIC■ – BIC■) | 1324.88 |
| Bayes Factor (BF■■) | 4.945195649273331e+287 |
| log■■ Bayes Factor | 287.69 |
| Posterior P(Model 1 \| 2025 data) | 100.0% |
| OLS Intercept (a) | -0.14 |
| OLS Slope (b) | 0.946 |
| Model 1 $R^2$ | 0.9564 |

### Verdict (Jeffreys' Scale): Decisive evidence for Model 1




Scatter of 2025 actual MB vs model-predicted xMB. Slope near 1 and intercept near 0 = good calibration.




Residual plot: xMB – MB vs actual MB for 2025.